

Christof Teuscher

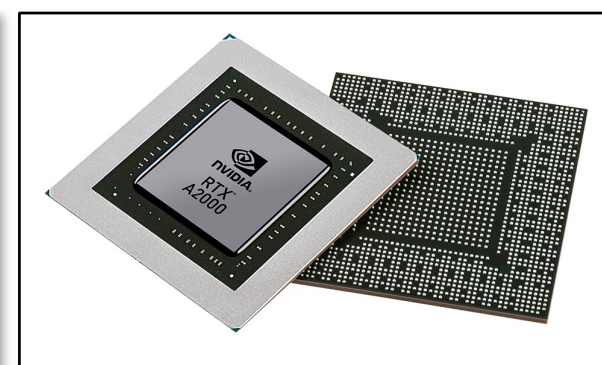
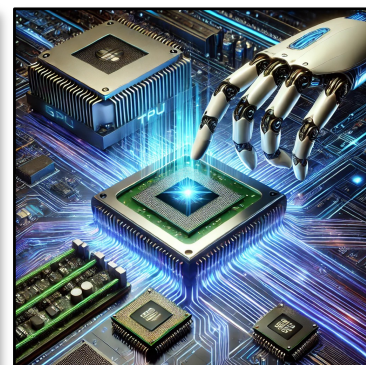
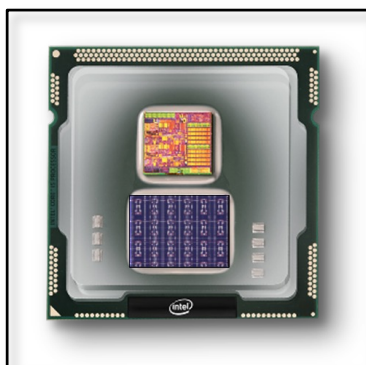
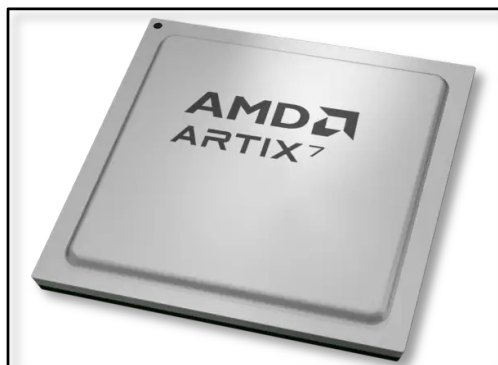
ECE 410/510: Hardware for AI and ML, Spring 2026

Week 3: Recap, outlook, and reminders

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What did you learn last week?

Can you spend \$250k on tokens per year?



Pradeep Ravindra • 2nd

AI dude in Biotech | Former U.S. Army Intelligence | Penn Data ...

4d •

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Jensen Huang said that a \$500k engineer is expected to spend \$250k in tokens per year.

Most people hear that and think - how the hell do you even spend that much in tokens?

You stop treating AI like a toy chatbot and start treating it like parallel engineering capacity.

The highest ROI for coding agents is:

- 1) Development loops
- 2) CI/CD [continuous integration/continuous operation]
- 3) Plan Mode

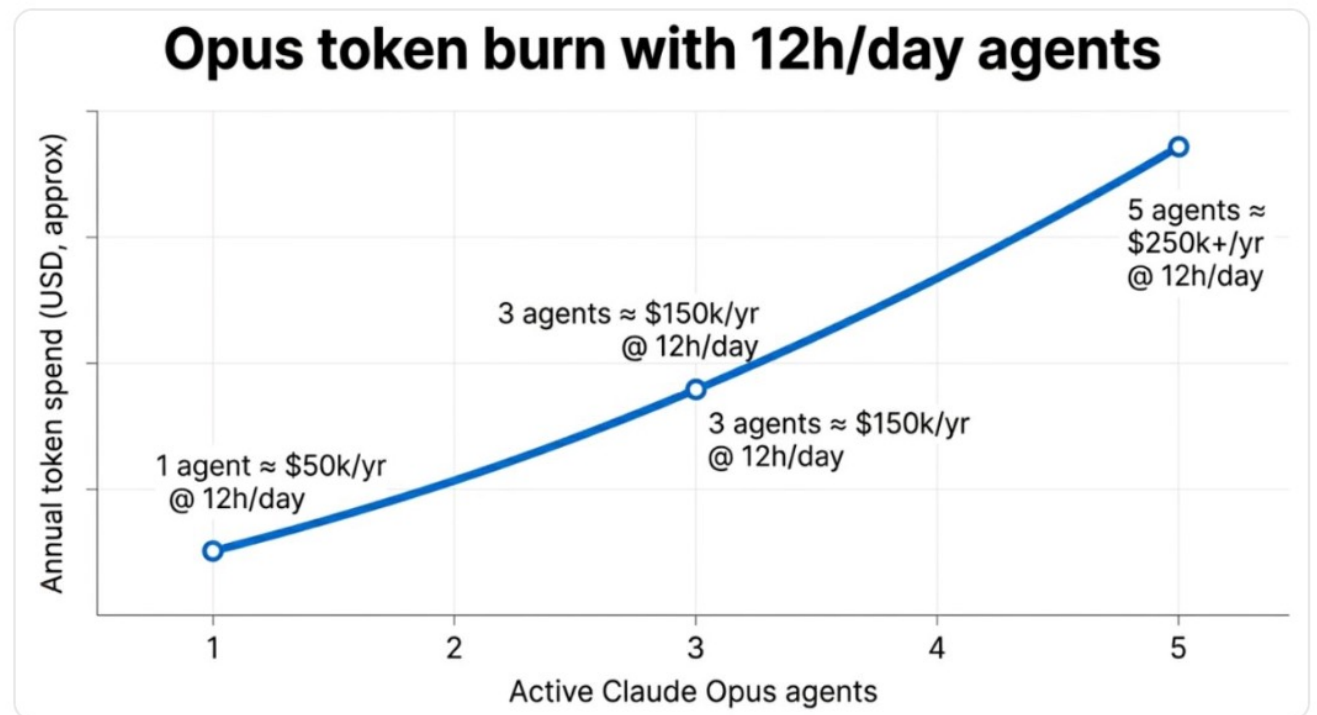
Now throw Git worktrees into the mix. One worktree per ticket, each with its own Claude Code session running in parallel.

Worktree A: feature branch

Worktree B: bugfix

Worktree C: migration in Plan Mode

Same engineer, multiple agents working at once. Suddenly, \$250k a year in tokens looks conservative.





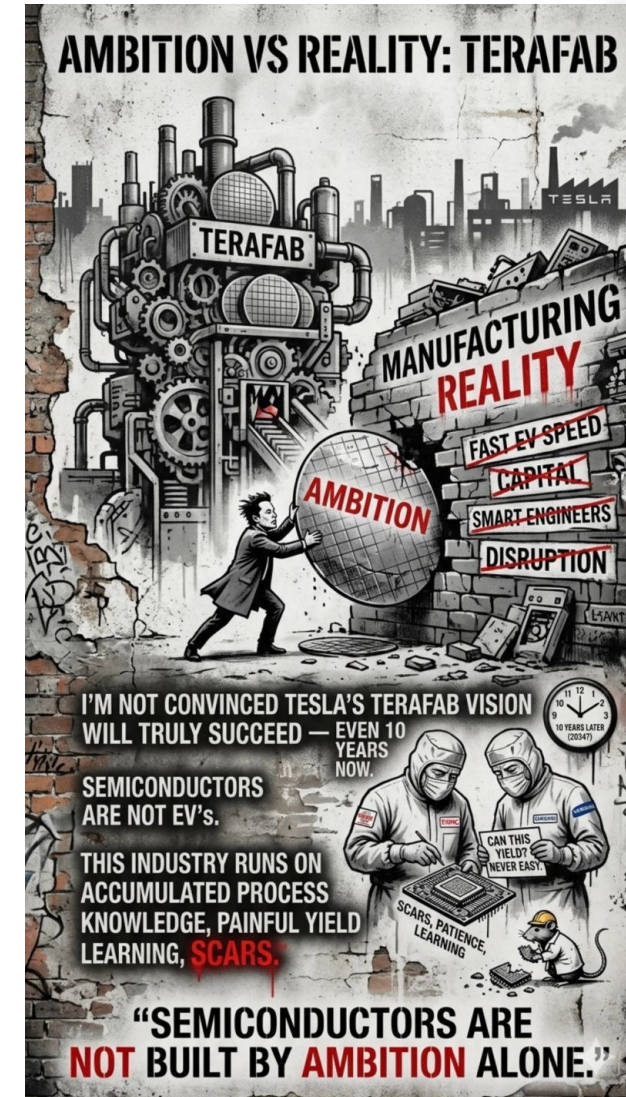
Updated 2 hours ago

Intel joins Musk's massive Terafab chipmaking project

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Edited by Emma W. Thorne, Editor at LinkedIn News

Intel [is joining](#) Terafab, the ambitious chipmaking project announced last month by Elon Musk. Shares in the California-based chip giant rose early Tuesday on the news. Terafab is a joint venture between Musk's companies SpaceX, xAI and Tesla that's meant to unite all the stages of the chipmaking process "[under one roof](#)," and [churn out chips](#) for Optimus robots and SpaceX satellites faster than other companies have been able to so far. Musk has touted the \$20 billion-plus project as "the most epic chip-building exercise in history by far."

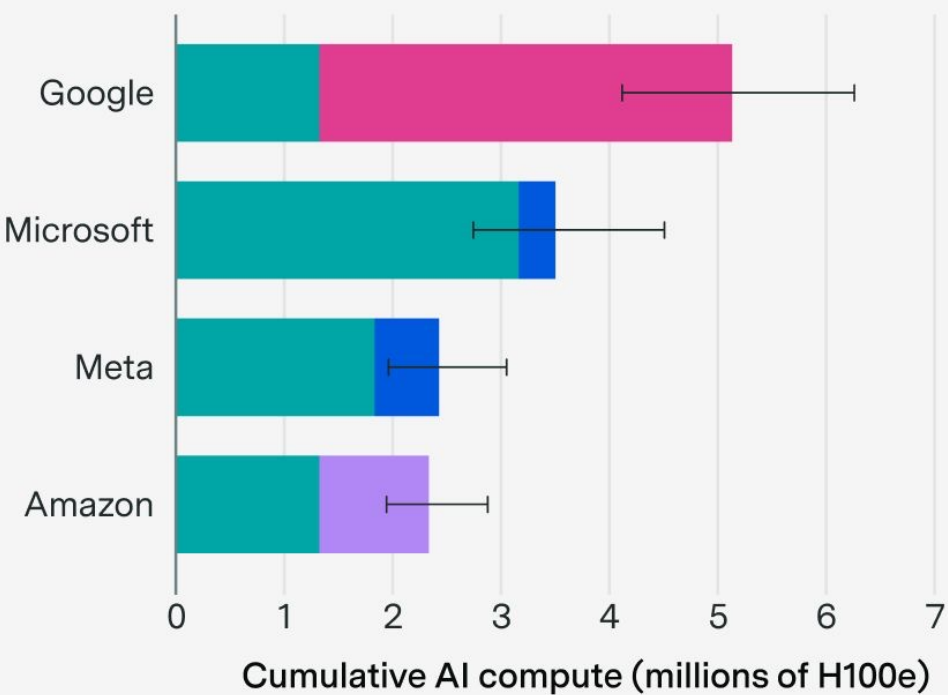


<https://www.linkedin.com/news/story/intel-joins-musks-massive-terafab-chipmaking-project-7160340>

Custom TPUs make Google the largest owner of AI compute

Cumulative chip acquired between Q1 2022 and Q4 2025, by chip manufacturer.

Nvidia Google TPU AMD AWS Trainium



Error bars correspond to 90% confidence intervals on total compute across all chip types.

Princeton researchers gave 12 AI models \$200K each and said: run a startup for a year. Hire people, find clients, manage cash, don't go bankrupt.

36 runs. A full year of simulation. Hundreds of business decisions in sequence. The results?

Only 5 out of 12 models turned a profit. The other 7 burned through everything. 47% of bankruptcies came down to one thing: the AI failed to detect an adversarial client.

Top 3 performers:

Claude Opus 4.6: \$1.27M in final funds. 0 bankruptcies in 3 runs. \$86 inference cost.

GLM-5: \$1.21M, also 0 bankruptcies. \$7.62 inference cost (11x cheaper).

GPT-5.4: over \$1M, 0 bankruptcies.

But the ranking isn't the interesting part. The decision-making patterns are. Models that survived focused on 1-2 key clients. Built trust. Deepened relationships. The ones that went bankrupt took on everyone, inflated headcount, and when 35% of clients turned out to be problematic, they had no resources left to recover.

Sound familiar? It's exactly what happens to real startups.

Here's what surprised me most: top-performing models kept a "scratchpad," writing out their reasoning before every decision. This turned out to be the single strongest predictor of survival. Not model size. Not speed. The habit of thinking before acting.

This is the first benchmark that measures not coding or math, but business reasoning over a long horizon. 3 out of 12 models turned \$200K into \$1M+. In one year.

AI can already run a business. Poorly, but it can. The question is who starts using it as a co-founder first.

Paper: arxiv.org/abs/2604.01212

[Submitted on 1 Apr 2026]

YC-Bench: Benchmarking AI Agents for Long-Term Planning and Consistent Execution

Muyu He, Adit Jain, Anand Kumar, Vincent Tu, Soumyadeep Bakshi, Sachin Patro, Nazneen Rajani

As LLM agents tackle increasingly complex tasks, a critical question is whether they can maintain strategic coherence over long horizons: planning under uncertainty, learning from delayed feedback, and adapting when early mistakes compound. We introduce YC-Bench, a benchmark that evaluates these capabilities by tasking an agent with running a simulated startup over a one-year horizon spanning hundreds of turns. The agent must manage employees, select task contracts, and maintain profitability in a partially observable environment where adversarial clients and growing payroll create compounding consequences for poor decisions. We evaluate 12 models, both proprietary and open source, across 3 seeds each. Only three models consistently surpass the starting capital of \$200K, with Claude Opus 4.6 achieving the highest average final funds at \$1.27 M, followed by GLM-5 at \$1.21 M at 11× lower inference cost. Scratchpad usage, the sole mechanism for persisting information across context truncation, is the strongest predictor of success, and adversarial client detection is the primary failure mode, accounting for 47% of bankruptcies. Our analysis reveals that frontier models still fail through distinct failure modes such as over-parallelization, demonstrating the capability gaps for long-horizon performance. YC-Bench is open-source, reproducible, and configurable.

<https://arxiv.org/abs/2604.01212>

Introducing Project Glasswing: an urgent initiative to help secure the world's most critical software.

It's powered by our newest frontier model, Claude Mythos Preview, which can find software vulnerabilities better than all but the most skilled humans: <https://lnkd.in/eS-fu9BF>

We've partnered with Amazon Web Services, Apple, Broadcom, Cisco, CrowdStrike, Google, JPMorganChase, the Linux Foundation, Microsoft, NVIDIA, and Palo Alto Networks. Together we'll use Mythos Preview to help find and fix flaws in the systems on which the world depends.

Mythos Preview has already found thousands of high-severity vulnerabilities—including some in every major operating system and web browser.

Given the pace of AI progress, it won't be long before models this capable are widespread. But there are strong reasons for optimism: AI will also be invaluable for defensive work.

We do not plan to make Mythos Preview generally available. Our goal is to deploy Mythos-class models safely at scale, but first we need safeguards that reliably block their most dangerous outputs. We'll begin testing those safeguards with an upcoming Claude Opus model.

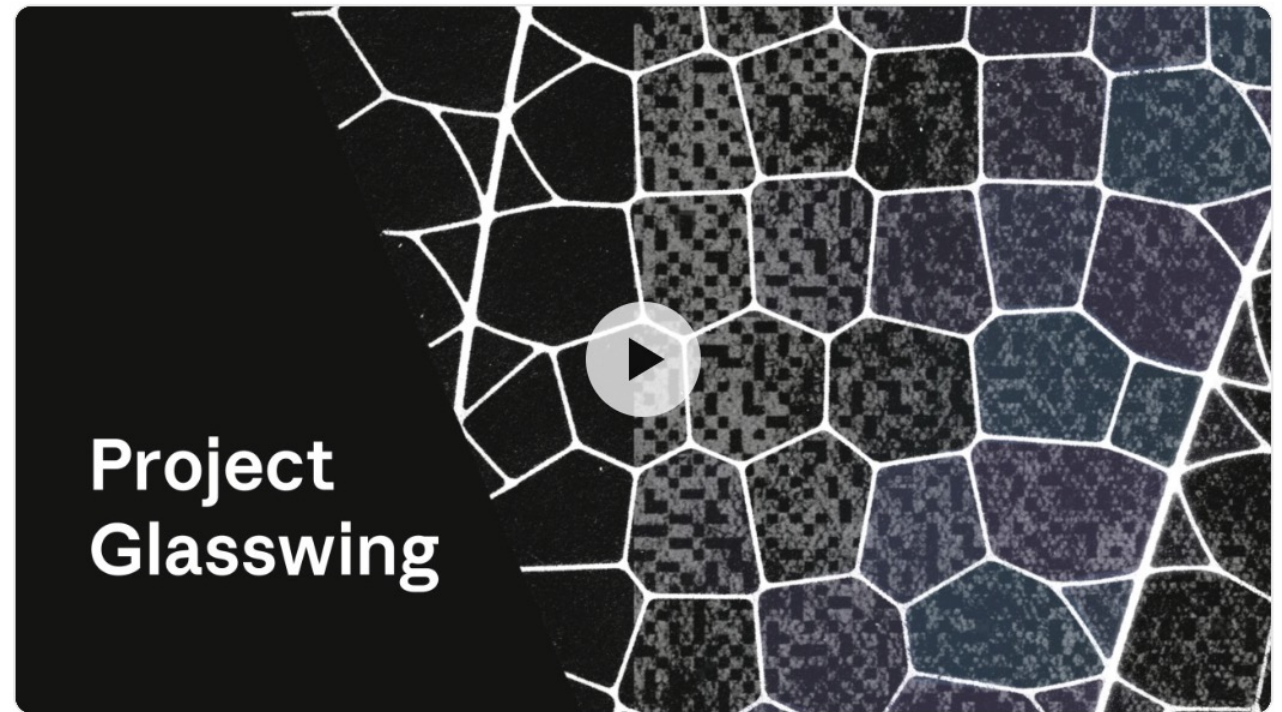
We're committing up to \$100M in Mythos Preview usage credits for our partners and over 40 other organizations that maintain critical software, including open-source projects. Anthropic will report back what we learn.

Project Glasswing is just a starting point. No organization can solve these cybersecurity problems alone: industry, open source, researchers, and governments all have essential roles to play.

You can read a detailed technical report on the software vulnerabilities and exploits discovered by Claude Mythos Preview here: <https://lnkd.in/eE7Kp9ZX>

The Claude Mythos Preview system card is available here: <https://lnkd.in/esH2-mfd>

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Mythos Preview vs GPT-5.4 Pro

Only four benchmarks are reported in common between Anthropic's Mythos Preview system card and the GPT-5.4 Pro column of OpenAI's launch post. Here's the head-to-head.

BENCHMARKS IN COMMON

GPQA Diamond	Mythos		94.5	≈ tie +0.1 MYTHOS
	5.4 Pro		94.4	
Humanity's Last Exam no tools	Mythos		56.8	+14.1 MYTHOS
	5.4 Pro		42.7	
Humanity's Last Exam with tools	Mythos		64.7	+6.0 MYTHOS
	5.4 Pro		58.7	
BrowseComp	Mythos		86.9	+2.4 5.4 PRO
	5.4 Pro		89.3	

API PRICING

	Mythos Preview	GPT - 5.4 Pro
Input	\$25 / MTok	\$30 / MTok
Output	\$125 / MTok	\$180 / MTok
Long context	same price	2× (\$60 in / \$270 out)
Access	Glasswing only	Public API

RPU – A Reasoning Processing Unit

Matthew Joseph Adiletta, Gu-Yeon Wei and David Brooks
Harvard University

Abstract—Large language model (LLM) inference performance is increasingly bottlenecked by the memory wall. While GPUs continue to scale raw compute throughput, they struggle to deliver scalable performance for memory bandwidth bound workloads. This challenge is amplified by emerging reasoning LLM applications, where long output sequences, low arithmetic intensity, and tight latency constraints demand significantly higher memory bandwidth. As a result, system utilization drops and energy per inference rises, highlighting the need for an optimized system architecture for scalable memory bandwidth.

To address these challenges we present the Reasoning Processing Unit (RPU), a chiplet-based architecture designed to address the challenges of the modern memory wall. RPU introduces: (1) A Capacity-Optimized High-Bandwidth Memory (HBM-CO) that trades capacity for lower energy and cost; (2) a scalable chiplet architecture featuring a bandwidth-first power and area provisioning design; and (3) a decoupled microarchitecture that separates memory, compute, and communication pipelines to sustain high bandwidth utilization. Simulation results show that RPU performs up to 45.3× lower latency and 18.6× higher throughput over an H100 system at ISO-TDP on Llama3-405B.

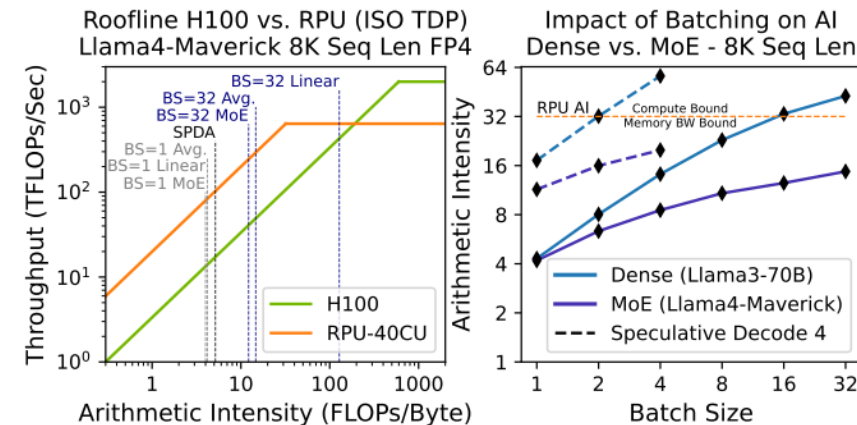


Fig. 1. RPU provides higher memory bandwidth than H100, which is required for low-latency decoding. Even up to BS=32, arithmetic intensity remains low, but requires the RPU to execute kernels which straddle the roofline.

System architects have responded to the decode bottleneck by cramming more HBM stacks into each package as the simplest way to chase bandwidth [14], [27], [61], [63]. How-

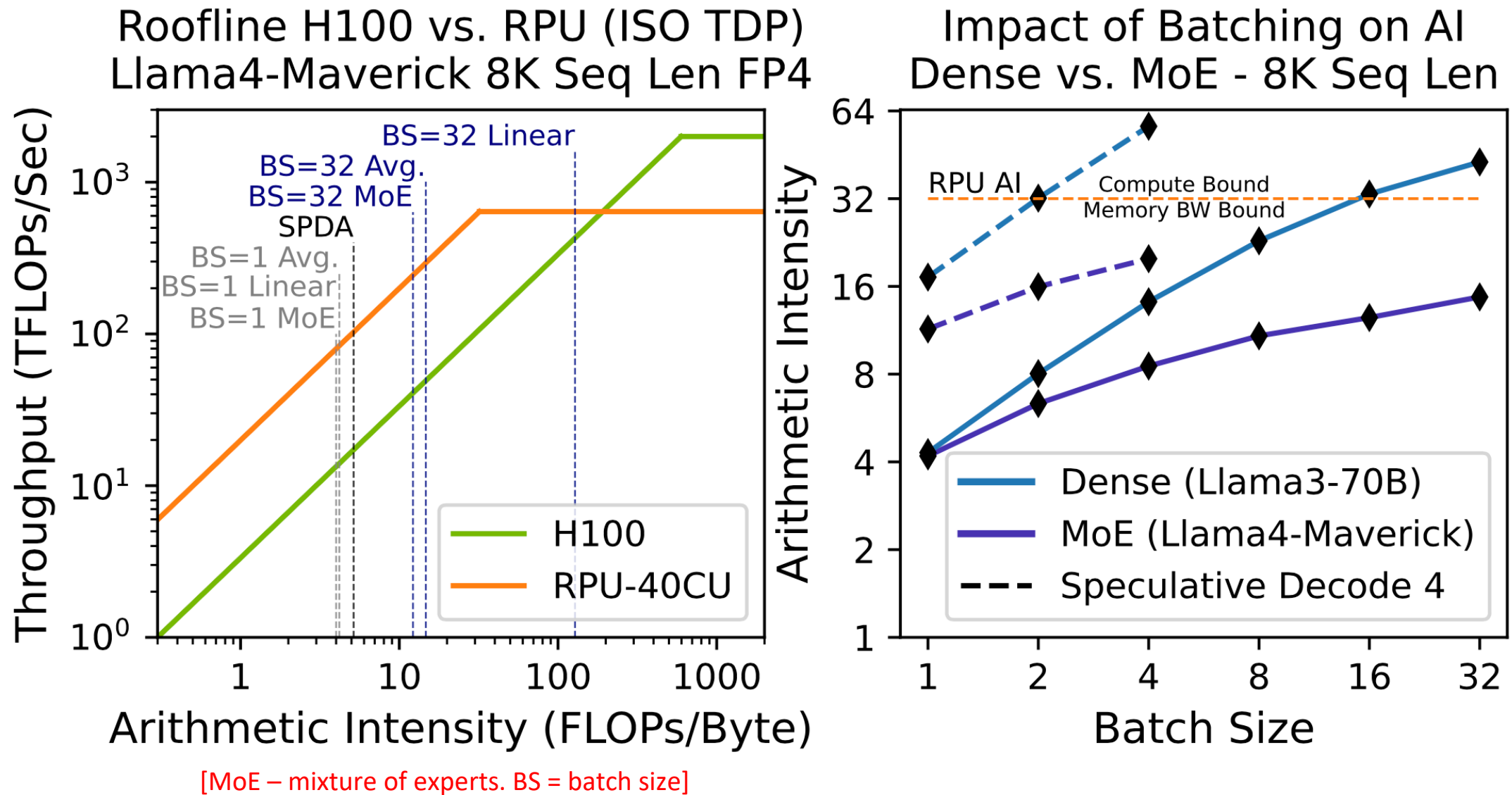


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Research Finds AI's Energy Use Is Driving Concern →

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News

[RDU = Reconfigurable Dataflow Unit]

SambaNova and Intel Announce Blueprint for Heterogeneous Inference: GPUs For Prefill, SambaNova RDUs for Decode, and Intel® Xeon® 6 CPUs for Agentic Tools

by SambaNova | [📅 April 8, 2026](#)

<https://sambanova.ai/press/sambanova-announces-collaboration-with-intel-on-ai-solution>

<https://sambanova.ai/products/rdu-ai-chips>

Landscape: Delivering Premium Inference on the Largest Workloads			
What systems are used for inference?	Prefill: GPU/RDU Agnostic Decode: SambaNova SN50 CPU: Intel Xeon 6	Prefill: Nvidia Rubin GPU Decode: Rubin GPU & Groq 3 LPU CPU: Vera CPU	Prefill: GPU Agnostic Decode: Cerebras CS-3 CPU: RISC-based CPU
How many decode chips are required to deliver Premium Inference on trillion parameter models?	256 Chips SambaNova SN50	2000+ Chips Groq 3 LPU	Not possible Cerebras CS-3
What data centers does this system fit in?	Plugs into existing data centers — requires just 30 kW per rack	Requires new liquid-cooled data center facility — 1 MW+	Not applicable
How well does the hardware adapt to changing premium workloads?	High utilization	Low utilization	Not applicable
What is the software coverage for agentic tools?	 Intel Xeon 6	 Nvidia Vera CPU	 RISC-based CPU

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AI Accelerator Market Looks Set to Exceed \$600 Billion by 2033, Driven by Hyperscale Spending and ASIC Adoption, According to Bloomberg Intelligence

GPU Market to Reach \$486 Billion; ASIC Market Heads to \$118 Billion

The research projects that GPUs will maintain their dominant position, capturing 81% of the AI accelerator market by 2033 with revenues growing at a 14% compound annual rate to reach approximately \$486 billion. Nvidia is expected to hold 70-75% of the AI accelerator market through 2030, with AMD positioned as the credible second source with a projected minimum 10% share by decade's end.

Meanwhile, custom ASICs are experiencing explosive growth, with the market projected to expand at a 27% compound annual rate to reach \$118 billion by 2033, up from just 8% of the total market in 2024 to 19% by 2033. Broadcom and Marvell are leading this transformation – Broadcom commanding 60-80% of the AI ASIC market through partnerships with Google, Meta and OpenAI, and Marvell holding a 20-25% share through key design wins with Amazon Web Services and Microsoft.